An octagon surgery explains all six correction diagrams

Finding. In each of the six representatives, square moves expose an alternating octagonal face with four hourglasses. A common cut-and-splice operation on this face, followed by four bivalent contractions, produces exactly the first correction diagram. In orbit 1 a second, ordinary-octagon unzip produces the second correction. This gives a geometric analogue of the benzene-chain picture, using an eight-sided configuration.

Notation and the sign distinction. Let R_i denote the unsigned web diagram reached by surgery, and let $W_i = W_{T_i}$. The earlier booklet's representative pages give the following formulas, with R_i abbreviating its web invariant. In particular, orbits 2 and 5 have a plus sign.

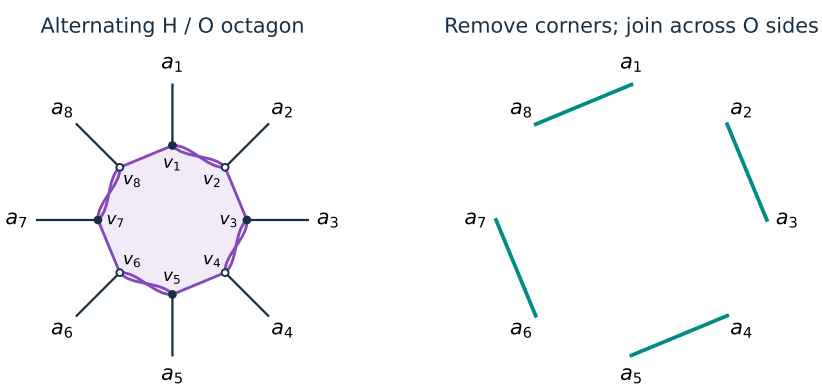
Individual orbit	T_i (row word)	Correction diagram R_i (row word)	Common-basis element
1	1111222233334444	1121323142324344	$C_1 = W_1 - R_1 - Q$
2	1111222233343444	1121323142342344	$C_2 = W_2 + R_2$
3	1111222323343444	1121323412342344	$C_3 = W_3 - R_3$
4	1112122233343444	1123123142342344	$C_4 = W_4 - R_4$
5	1112122323343444	1123123412342344	$C_5 = W_5 + R_5$
6	1121223133424344	1231234123412344	$C_6 = W_6 - R_6$

Here $Q = W_{1212341234123434}$. To use $C_i = W_i - P_i$ uniformly for $i = 2, \ldots, 6$, set $P_2 = -R_2$, $P_5 = -R_5$, and $P_i = R_i$ otherwise.

What has been checked. The output diagrams agree with the specified correction JSONs, fixing all sixteen boundary labels, vertex colors, edge multiplicities, and cyclic orders. Both exposed-octagon presentations in each complete square orbit give the same specified correction. No benzene face occurs in any of the 176 presentations enumerated.

Scope. This is a verified *geometric rule selecting correction diagrams*. It is not a skein identity $W_i = R_i$, and does not derive the signs or coefficients in the displayed C_i formulas. Those are inherited from the booklet, not proved by this surgery. Theta membership is not asserted.

The local operation: splice across the ordinary sides



Write the exposed octagon cyclically as $v_1, \ldots, v_8$, with hourglass sides $v_1v_2, v_3v_4, v_5v_6, v_7v_8$. Each corner has one exterior ordinary edge v_ja_j. The ordinary sides are the other four sides.

1. Delete the eight corners and their incident edges.
2. Reconnect the exterior stubs across the ordinary sides:

$$a_2a_3, \quad a_4a_5, \quad a_6a_7, \quad a_8a_1.$$

3. In these six examples, each new edge duplicates an existing ordinary edge. Replace each such pair by a multiplicity-2 *hourglass*, retaining the transported cyclic positions.
4. Contract the four resulting bivalent 2+2 vertices, merging their same-color neighbors. The result is R_i.

Normalization is part of the new operation. Step 3 is a prescribed replacement by an exterior-square edge with its hourglass twist. It is not an assertion that two ordinary tensor strands equal an exterior-square strand with coefficient one. The final bivalent contractions are the familiar hourglass-web contraction moves; the whole splice is a new surgery, not an existing invariant-preserving move.

Only the initial square moves preserve the invariant represented by the diagram: $W_i = \widetilde{W}_i$ under the established square-move convention [1, Figures 2 and 9; Section 6.3]. The surgery then changes the web to R_i. In symbols, with different arrows for these different roles,

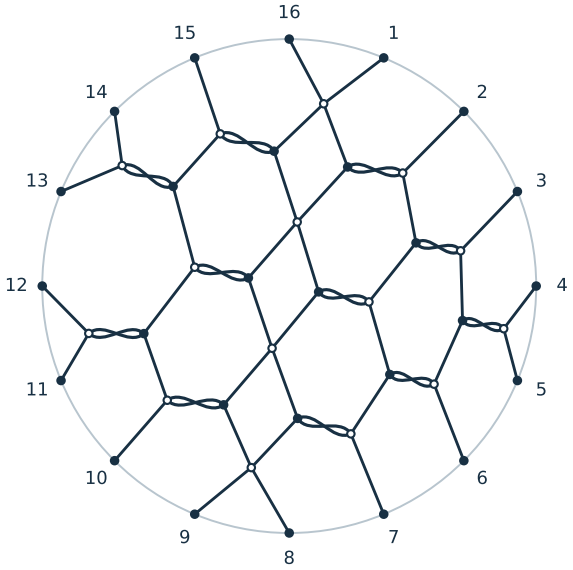
$$W_i \overset{\text{square moves}}{=} \widetilde{W}_i \xrightarrow{\text{ordinary-side octagon surgery}} R_i.$$

The operation resembles benzene surgery by selecting alternating sides of an even face, but here the face has eight sides and four hourglasses, not six sides and three hourglasses.

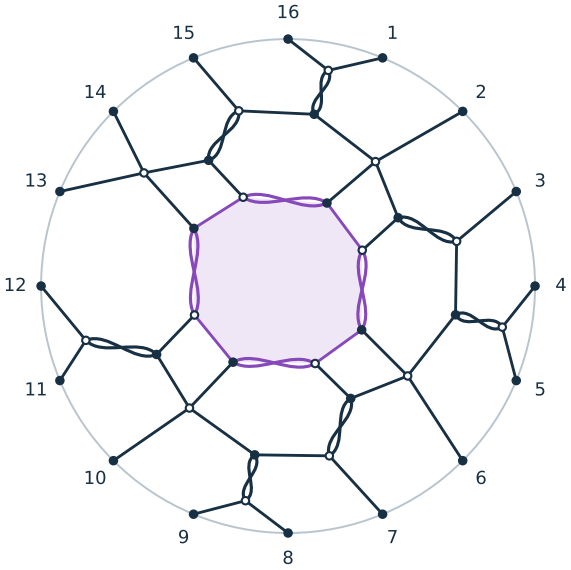
Individual orbit 2

Orbit 2. $T_2 = 1111222233343444$, $R_2 = W_{1121323142342344}$, $\boxed{C_2 = W_2 + R_2}$.

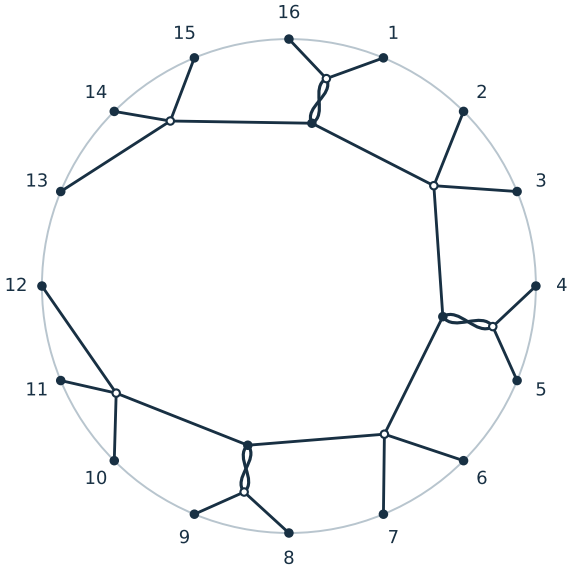
Original W_T



$\widetilde{W}_T$: exposed octagon



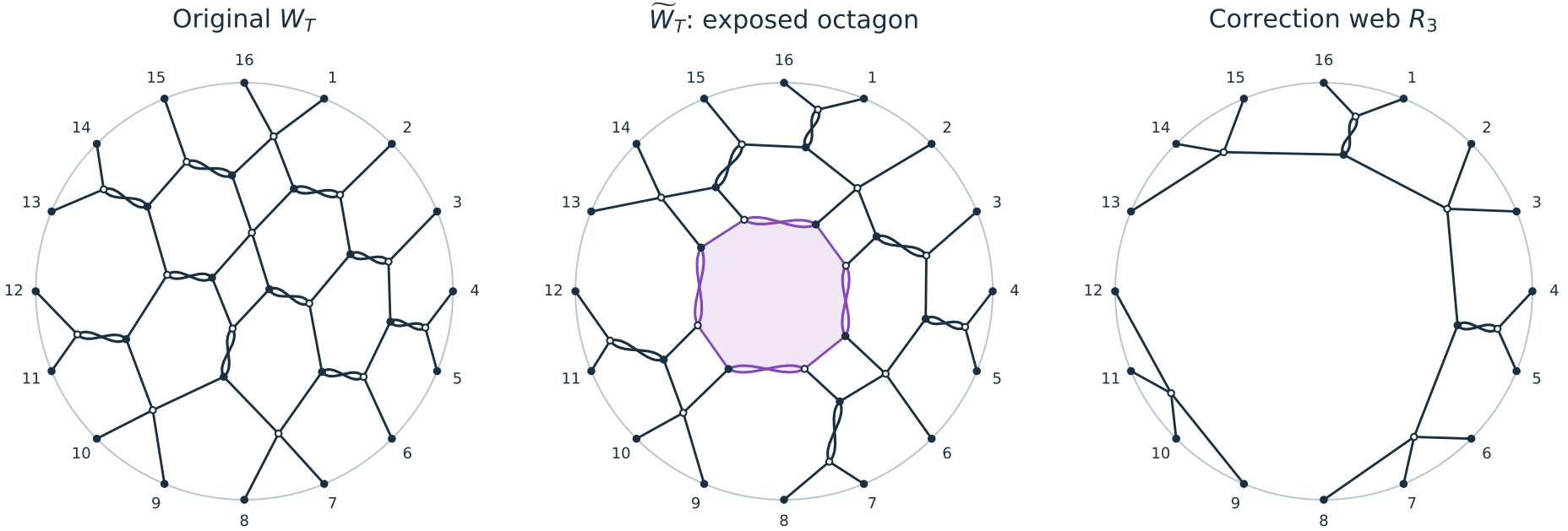
Correction web R_2



The middle presentation is reached by five square moves. Splicing the four ordinary octagon sides and making four $2+2$ contractions gives the right-hand diagram. In the notation $C_2 = W_2 - P_2$, the signed correction is $P_2 = -R_2$.

Individual orbit 3

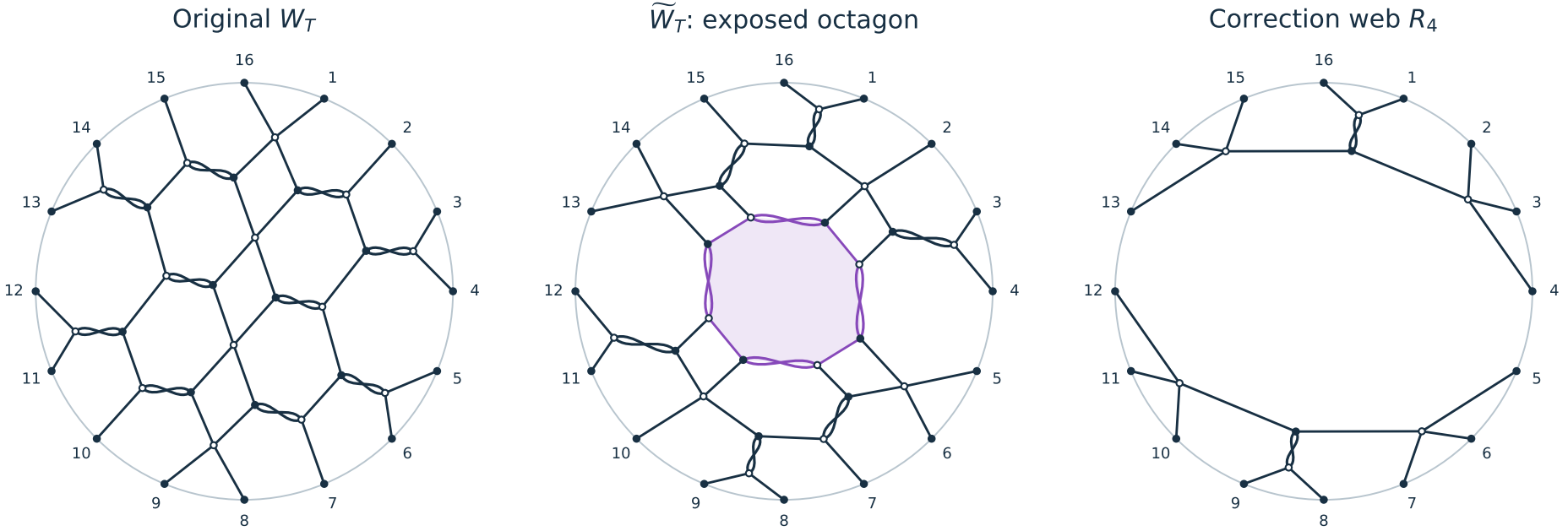
Orbit 3. $T_3 = 1111222323343444$, $R_3 = W_{1121323412342344}$, $\boxed{C_3 = W_3 - R_3}$.



Four square moves expose the same local octagon type. The identical surgery gives R_3. The boundary labeling is unchanged throughout.

Individual orbit 4

Orbit 4. $T_4 = 1112122233343444$, $R_4 = W_{1123123142342344}$, $\boxed{C_4 = W_4 - R_4}$.

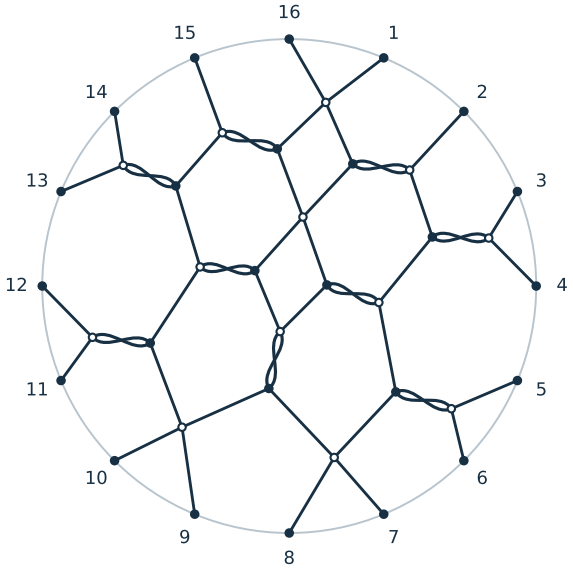


Five square moves expose the alternating octagon. After the same splice and four contractions, the correction has eight internal vertices and two hourglasses.

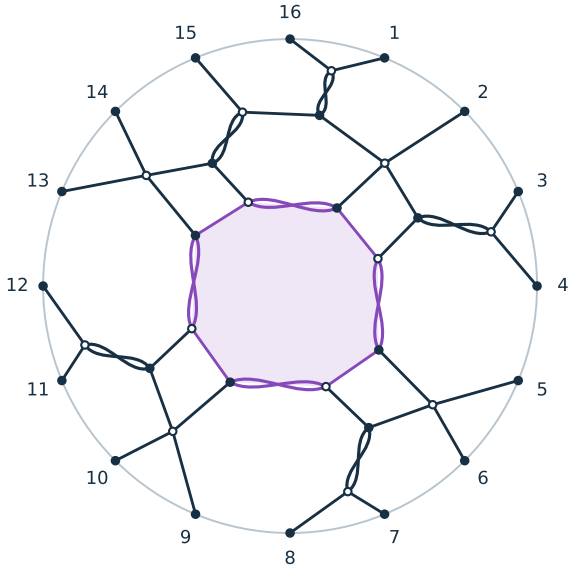
Individual orbit 5

Orbit 5. $T_5 = 1112122323343444$, $R_5 = W_{1123123412342344}$, $\boxed{C_5 = W_5 + R_5}$.

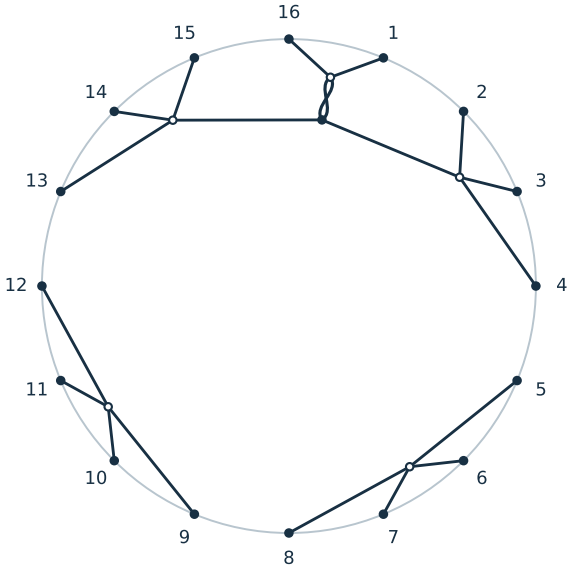
Original W_T



$\widetilde{W}_T$: exposed octagon



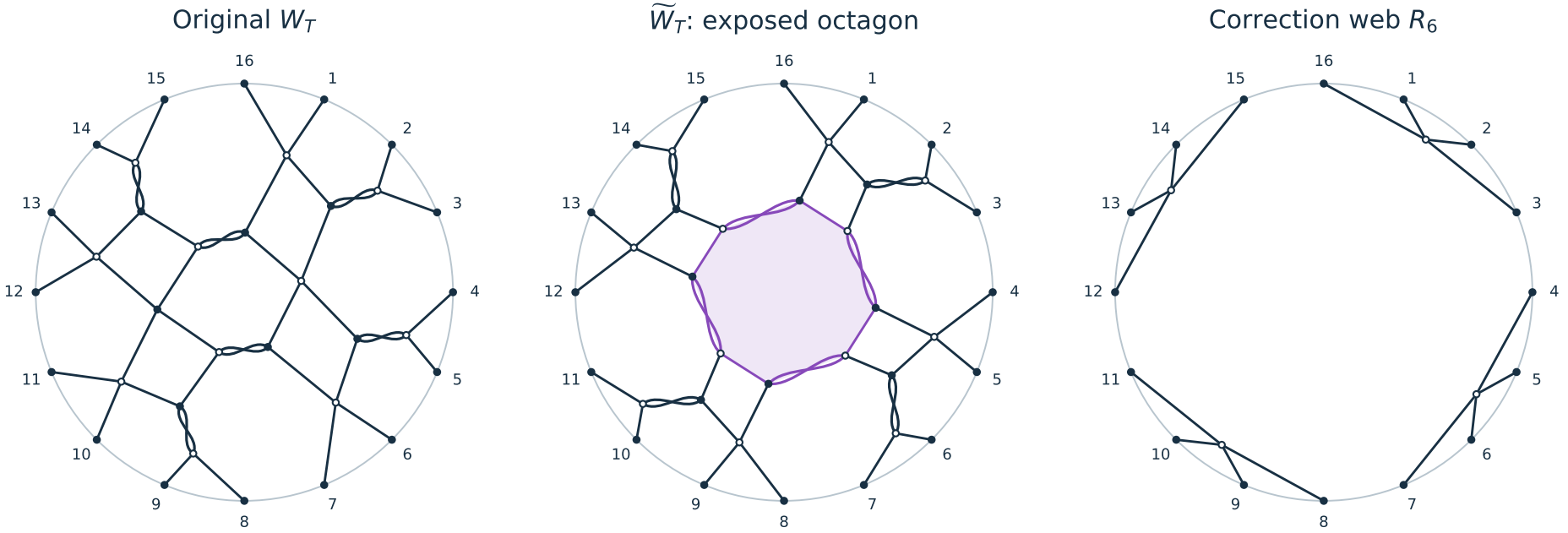
Correction web R_5



Four square moves expose the octagon. The output has six internal vertices and one hourglass. Again the geometry produces R_5, while the booklet's signed correction is $P_5 = -R_5$.

Individual orbit 6: the correction becomes four stars

$$T_6 = 1121223133424344, \qquad R_6 = W_{1231234123412344}, \qquad \boxed{C_6 = W_6 - R_6}.$$



Two square moves suffice. The exposed diagram has twenty internal vertices and eight hourglasses; removing the octagon and performing the four bivalent contractions leaves four internal vertices and no hourglasses.

The four star components have the following boundary blocks:

$$\{1,2,3,16\}, \quad \{4,5,6,7\}, \quad \{8,9,10,11\}, \quad \{12,13,14,15\}.$$

Consequently, as a product of Plücker coordinates, this correction is proportional, with the orientation normalization determined by its source JSON, to

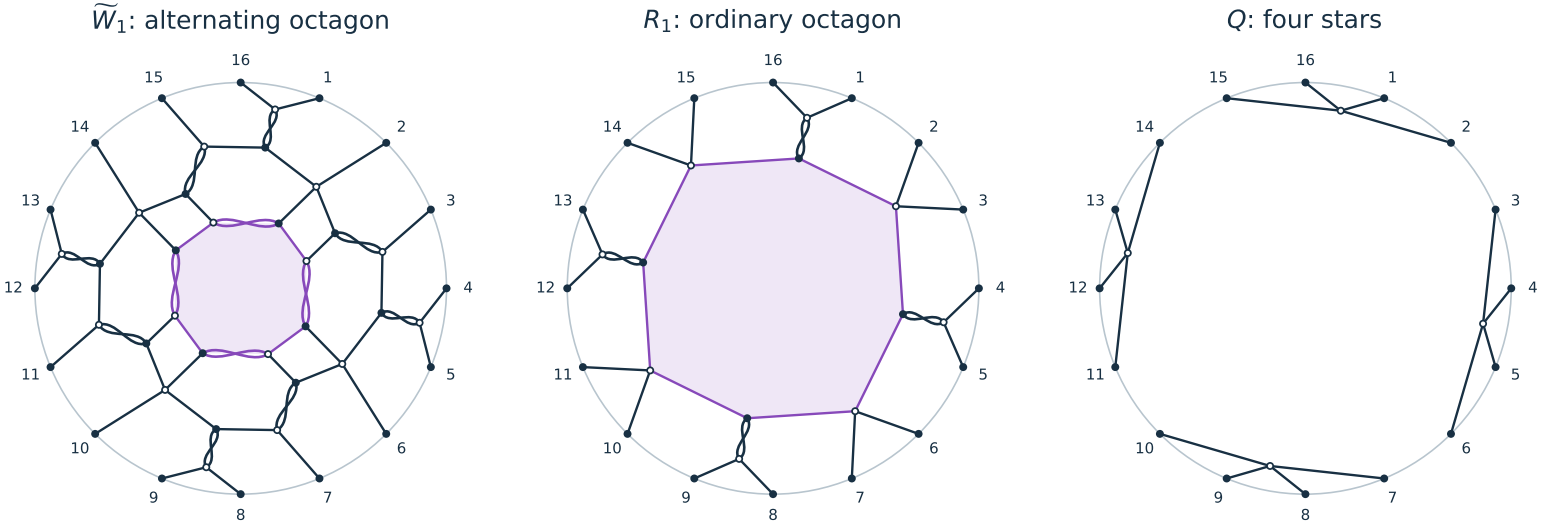
$$\Delta_{1,2,3,16}\,\Delta_{4,5,6,7}\,\Delta_{8,9,10,11}\,\Delta_{12,13,14,15}.$$

All four factors are cyclic interval Plücker coordinates. This identifies the shape of the correction without using any theta-basis inference.

A useful visual interpretation. The surgery removes a central alternating resonance region and leaves its exterior attachments to determine the smaller web. Thus the correction need not arise from changing an isolated benzene in the original drawing: it can be revealed only after square moves.

Individual orbit 1: an octagon chain

$$C_1 = W_{1111222233334444} - W_{1121323142324344} - W_{1212341234123434}$$



First stage. Five square moves expose the left-hand alternating octagon. The ordinary-side splice and four contractions produce R_1, the middle diagram. It has twelve internal vertices and four hourglasses.

Second stage: unzip. The middle diagram contains an ordinary octagon. Alternating corners have either one exterior hourglass or two exterior ordinary edges. Delete its eight vertices; split each exterior hourglass into two ordinary attachments and connect them to the adjacent ordinary stubs, one on each neighboring side of the face. Untwist the pair at the surviving exterior endpoint. This last order convention is essential: merely forgetting the twist gives a different ribbon graph.

More explicitly, write the octagon as $h_1, u_1, h_2, u_2, h_3, u_3, h_4, u_4$. Let h_i have its exterior hourglass to z_i. Let the exterior ordinary stubs at u_i lead to a_i on the h_i side and b_i on the h_{i+1} side. Replace the octagon and spokes by

$$z_i a_i \quad \text{and} \quad z_i b_{i-1} \qquad (i \bmod 4),$$

with the order at z_i reversed relative to the removed hourglass pair. The result is precisely Q, whose star blocks are

$$\{1,2,15,16\}, \quad \{3,4,5,6\}, \quad \{7,8,9,10\}, \quad \{11,12,13,14\}.$$

The geometric chain is $\widetilde{W}_1 \rightsquigarrow R_1 \rightsquigarrow Q$; C_1 subtracts both descendants.

Verification, reproducibility, and the remaining algebraic question

Orbit	Square states	Exposing moves	Octagon presentations	Exposed internal/HG	Output internal/HG
1	42	5	2	28 / 12	12 / 4
2	35	5	2	26 / 11	10 / 3
3	28	4	2	24 / 10	8 / 2
4	30	5	2	24 / 10	8 / 2
5	23	4	2	22 / 9	6 / 1
6	18	2	2	20 / 8	4 / 0

Exact checks. Breadth-first search enumerated each complete square orbit, checking the three trip permutations after every move. For every exposed alternating octagon, the splice produces four doubled pairs and four bivalent $2 + 2$ contractions. A boundary-fixed isomorphism to the correction JSON preserves colors, multiplicities and all compressed cyclic orders, with each multiplicity-2 edge understood as the standard hourglass. All twelve checks pass. The second-stage unzip also matches Q with these data. These checks are combinatorial and independent of the drawing layout.

Files. The source bundle contains the LaTeX, vector illustrations, original graph JSONs, generated JSONs for exposed/spliced/contracted diagrams, the complete matching certificates, square-move paths, and replay code. The generated schema stores `rotation_ccw` as a cyclic list of neighbors; multiplicity-2 neighbors occur once in this compressed list. Original JSONs retain their full strand data. Figure coordinates are for display only; topology is taken from the stored rotation systems.

What remains to prove. To promote this geometric explanation to an algebraic construction of the common basis, one must derive the signed coefficients from the FLL pairing or from an appropriate local relation. A general surgery rule would also need a proof of compatibility with arbitrary square-move presentations. The present finite check establishes consistency on these six entire square orbits, not on all webs in $\mathrm{Gr}(4, 16)$.

References

[1] C. Gaetz, O. Pechenik, S. Pfannerer, J. Striker, and J. P. Swanson, *Rotation-invariant web bases from hourglass plabic graphs*, arXiv:2306.12501. Figures 2 and 9 and Section 6.3 supply the square-move and contraction context; the octagon correction surgeries here are computational observations.

[2] *Common_C_basis_68_cases_and_theta_tests.pdf*, individual orbit representative pages 5–10. Source of the six common-basis formulas and their sign conventions.

[3] Z. Saron, *Tableaux–Hourglass Plabic Graphs and Web Duality*, project repository. The supplied graph JSONs and the earlier booklet, rather than a new pairing calculation, are the inputs to this comparison.